A Defect–Cokernel Duality for a Restricted Table-of-Marks Map

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Abstract

The preceding exact calculations give a restricted 16×16 self-mark matrix M and a three-dimensional C65 saturation defect. We identify the embedded defect subgroup inside the C66 cokernel and compute the quotient on both column and row sides. The defect subgroup is $(\mathbb{Z}/8) \oplus (\mathbb{Z}/2)^2$, while the quotient has invariant factors $(1^4, 2^8, 4^2, 12, 144)$. An explicit transpose-side congruence lattice gives the dual quotient with the same invariants. Every statement concerns only the frozen sixteen subgroup types.

1 Question and scope

Let $G = W(E_6)$ and let $S_1, \ldots, S_{16}$ be the fixed subgroup types from the preceding source package. Write

$$M_{ij} = |(G/S_j)^{S_i}|, \qquad M \in \mathrm{Mat}_{16}(\mathbb{Z}).$$

The matrix has determinant $226492416 = 2^{23}3^3$ and is the exact C64 restricted self-mark map. C65 supplies three integer kernel vectors

$$z_1 = S_{10} - S_9, \quad z_2 = -S_2 - S_3 - S_5 - S_6 + S_{11} + S_{12} + S_{13} + S_{14}, \quad z_3 = S_{16} - S_{15}.$$

Put $C = \mathbb{Z}^{16}/M\mathbb{Z}^{16}$. C66 computes the ambient Smith invariants, while C67 computes coordinate-wise inverse denominators. C68 asks how the C65 saturation vectors sit inside C and what the transpose-side annihilator looks like.

The scope firewall is

`NO_BAD_EULER_OR_ROOT_NUMBER`.

We do not claim a full table of marks, a full Burnside ring, arithmetic resolvents, local fields, Euler factors, root numbers, automorphy, or a Hilbert–Polya operator.

2 Defect subgroup in the cokernel

Let u_1, u_2, u_3 be the C65 saturation vectors. Direct exact multiplication gives

$$Mz_1 = 8u_1, \qquad Mz_2 = 2u_2, \qquad Mz_3 = 2u_3.$$

Thus the classes $[u_i]$ are torsion classes in C. The least multipliers are detected by the exact rational matrix $M^{-1}U$, where $U = [u_1\ u_2\ u_3]$.

Theorem 1 (Embedded defect and quotient)**.** *Let $D = \langle [u_1], [u_2], [u_3] \rangle \subseteq C$. Then*

$$D \cong \mathbb{Z}/8 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/2,$$

and

$$C/D \cong (\mathbb{Z}/2)^8 \oplus (\mathbb{Z}/4)^2 \oplus \mathbb{Z}/12 \oplus \mathbb{Z}/144.$$

Equivalently, the Smith invariants of the augmented matrix $[M \mid U]$ are

$$(1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 4, 4, 12, 144).$$

Proof. The three displayed relations show that the classes have orders dividing $8, 2, 2$. Exact rational inversion shows that the corresponding columns of $M^{-1}U$ have least denominators $8, 2, 2$, and enumeration of the $8 \cdot 2 \cdot 2$ coefficient classes shows that no nonzero combination is integral. Hence $|D| = 32$ and its invariant factors are $(2, 2, 8)$. Independently, Euclidean Smith reduction of $[M \mid U]$ gives the displayed diagonal. Its product is $7077888 = 226492416/32$, as required for C/D. $\square$

3 Transpose-side annihilator

Define the congruence lattice

$$A = \{y \in \mathbb{Z}^{16} : z_1^T y \equiv 0 \pmod 8,\ z_2^T y \equiv 0 \pmod 2,\ z_3^T y \equiv 0 \pmod 2\}.$$

The three functionals are well-defined on the transpose quotient because $z_i^T M^T = d_i u_i^T$ with $(d_1, d_2, d_3) = (8, 2, 2)$. In the named coordinate order, the residue rows $([z_1^T e_j]_8, [z_2^T e_j]_2, [z_3^T e_j]_2)$ are

j	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
residue	000	010	010	000	010	010	000	000	700	100	010	010	010	010	001	001

so the coordinate types annihilated individually are S_1, S_4, S_7, S_8.

Proposition 1 (Row–column duality). *The lattice A contains $M^T\mathbb{Z}^{16}$ with index 32, and*

$$A/M^T\mathbb{Z}^{16} \cong (C/D)^\vee.$$

Its Smith invariants are $(1^4, 2^8, 4^2, 12, 144)$, identical to those of C/D.

Proof. An explicit integral basis P for A has determinant 32. Multiplying $P^{-1}M^T$ gives an integer matrix with Smith diagonal $(1^4, 2^8, 4^2, 12, 144)$. The congruence equations define the annihilator of D under the natural pairing between the column and transpose cokernels; therefore the row quotient is the Pontryagin dual of C/D. □

4 Reproducibility and limitations

The evidence binds C64, C65, C66, and C67 source bytes by SHA-256. The producer and checker use separate exact Smith reductions; a SymPy integer Smith calculation reproduces both quotient diagonals. Clean replay passes, and seventeen hostile semantic mutations are rejected. No Smith transformation basis is treated as canonical. The theorem remains a finite integral statement on the frozen 16-type support.

5 Conclusion

C68 locates the C65 dyadic defect inside the C66 cokernel rather than merely reporting either group in isolation. The explicit row congruence lattice supplies the matching annihilator and makes the column/row duality checkable without additional representation-theoretic or arithmetic assumptions.